

b-secure User Mobile App

The b-secure User App allows customers to book licensed security guards on-demand or on a scheduled basis. Users can track guards in real-time on a map, make payments via Razorpay, manage a wallet, and trigger an SOS alert during active bookings.

Tech Stack

Layer	Technology
Framework	React Native via Expo SDK 51
Language	TypeScript (strict mode)
Navigation	Expo Router v3 (file-based routing)
Server State	TanStack Query v5 (React Query)
Client State	Zustand
Maps	React Native Maps + Google Maps SDK
Real-time	WebSocket (native browser API)
Payments	Razorpay React Native SDK
Push Notifications	Expo Notifications
Device Location	Expo Location
Styling	NativeWind v4 (Tailwind CSS for RN)
Storage	Expo SecureStore (JWT) + AsyncStorage
Error Tracking	Sentry (@sentry/react-native)
HTTP Client	Axios
Forms	React Hook Form + Zod

Folder Structure

```
bsecure-user-app/  
├─ app/                # Expo Router – all screens live  
here  
├─ _layout.tsx         # Root layout: auth gate,  
providers
```

```

├── (auth)/
│   ├── _layout.tsx      # Auth stack layout
│   ├── login.tsx        # Phone number entry
│   └── otp-verify.tsx    # OTP verification
├── (tabs)/
│   ├── _layout.tsx      # Bottom tab navigator
│   └── home.tsx         # Map + nearby guards + booking
entry
├── bookings.tsx         # Active & history bookings
├── wallet.tsx           # Wallet balance & top-up
├── profile.tsx          # User profile & settings
├── guards/
│   └── [id].tsx         # Guard detail / profile
├── booking/
│   ├── create.tsx       # Multi-step booking creation
│   ├── tracking.tsx     # Live tracking during booking
│   └── [id].tsx         # Booking detail / receipt
└── sos.tsx             # SOS hold-to-trigger screen

├── src/
│   └── api/             # Axios instance + all service
modules
├── axios.ts
├── authService.ts
├── guardService.ts
├── bookingService.ts
├── walletService.ts
├── reviewService.ts
├── userService.ts
├── sosService.ts
├── notificationService.ts
├── components/         # Reusable UI components
│   ├── GuardCard.tsx
│   ├── BookingCard.tsx
│   ├── MapGuardMarker.tsx
│   ├── OTPInput.tsx
│   ├── RatingStars.tsx
│   ├── StatusBadge.tsx
│   ├── SOSButton.tsx
│   └── LoadingOverlay.tsx
├── hooks/              # Custom React hooks
│   ├── useNearbyGuards.ts
│   ├── useCreateBooking.ts
│   ├── useBookingWebSocket.ts
│   ├── useWallet.ts
│   ├── useRazorpay.ts
│   ├── useLocation.ts
│   ├── useOTP.ts
│   ├── usePushNotifications.ts
│   └── useSOSTrigger.ts
└── store/              # Zustand global state

```

- useAuthStore.ts - useBookingStore.ts - useLocationStore.ts	
- types/ - index.ts	# TypeScript interfaces & enums
- lib/ - queryClient.ts - constants.ts	# TanStack Query configuration # App-wide constants
- utils/ - format.ts - validators.ts	# Date, currency formatters # Zod schemas
- assets/ - images/ - fonts/	
app.json	# Expo config
babel.config.js	
metro.config.js	
tailwind.config.js	
tsconfig.json	
package.json	

Documentation

File	Description
overview.md	Setup, environment variables, global state, Axios, React Query, TypeScript types
screens.md	Every screen with full TSX component code
components.md	Reusable UI components with typed props
hooks.md	All custom hooks with full implementations
api_integration.md	API service layer, interceptors, error handling

Quick Start

1. Clone and install

```
git clone https://github.com/bsecure/user-app.git
cd user-app
npm install
```

2. Set up environment (see overview.md)

```
cp .env.example .env
# fill in API_BASE_URL, WS_BASE_URL, GOOGLE_MAPS_API_KEY,
  RAZORPAY_KEY_ID, SENTRY_DSN
```

3. Start development server

```
npx expo start
```

4. Run on device/simulator

```
npx expo run:ios
npx expo run:android
```

Related Repositories

- bsecure-backend — Django REST Framework + Django Channels API
- bsecure-guard-app — Guard-side React Native app
- bsecure-admin — Web dashboard (Next.js)